What Drives Smartphone Prices?

ETW2500 Assignment 2

Presenters

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Introduction

Dataset and business question

What drives smartphone prices in the Indian market: overall specifications, specification trade-offs or brand?

- **980 smartphone models/variants**, 25 original variables.
- **Indian market**, with prices retained in **Indian rupees (₹)**.
- **Business context:** pricing and competitive positioning.
- **Kaggle dataset**, with the full citation on the References slide near the end.

PCA input variables

Variable	Type	Meaning / unit
processor_speed	Numeric	Processor speed (GHz)
ram_capacity	Numeric	RAM (GB)
internal_memory	Numeric	Internal storage (GB)
refresh_rate	Numeric	Display refresh rate (Hz)
primary_camera_rear	Numeric	Main rear-camera resolution (MP)
primary_camera_front	Numeric	Main front-camera resolution (MP)

Table 1: Specifications used to build the principal components

- **Six inputs**, standardised because their units differ (Table 1).
- Megapixels measure **resolution**, not overall image quality.

Outcome and extra predictors

Variable	Type	Meaning / role
price	Numeric	Listed price (₹), used to create the outcome
log_price	Numeric	Natural log of price in ₹, the outcome
battery_capacity	Numeric	Battery capacity (mAh)
screen_size	Numeric	Screen diagonal (inches)
has_5g	Categorical	5G available, No / Yes
has_nfc	Categorical	NFC available, No / Yes
brand_group	Categorical	Top seven brands plus Other, Xiaomi reference

Table 2: Outcome and extra predictors (not used in the PCA)

- **Five extra predictors:** two numeric and three categorical (Table 2).
- Price and these extra predictors are **excluded from PCA construction**.

Why this dataset suits PCA

- **Business relevance:** compare specification bundles and product positioning.
- **Shared information:** RAM and storage correlate at **0.65** in the complete records.
- **Suitable structure:** six numeric PCA inputs, a numeric outcome, and numeric and categorical extra predictors.
- **Usable sample: 921 complete records**, with **689** used to fit PCA.

Objectives and significance

Objectives

1. **Reduce** six related smartphone specifications to a few principal components and **interpret** what each component represents.
2. **Assess** how the components, battery capacity, screen size, 5G, NFC and brand are associated with smartphone price.

Significance

- **Manufacturers:** identify which specification bundles support a higher price when designing new models.
- **Retailers:** set price ranges and plan product assortments by brand and features.
- **Consumers:** judge whether a higher price reflects better specifications or only the brand.

Descriptive Analysis

Missing values and cleaning

Variable	Missing values
Processor speed (GHz)	42
Front camera (MP)	5
Battery capacity (mAh)	11

Table 3: Missing values after removing the two luxury editions

- **3 used variables** have missing values, with processor speed having the most (Table 3).
- **2 luxury editions** excluded to match the group's product scope.
- **57 incomplete records** removed, leaving **921 phones**.

Descriptive summary: PCA inputs

Specification	Mean	SD	Median	Min	Max
Processor speed (GHz)	2.42	0.46	2.30	1.20	3.22
RAM (GB)	6.56	2.74	6.00	1.00	18.00
Storage (GB)	137.42	97.22	128.00	8.00	1,024.00
Refresh rate (Hz)	92.59	28.82	90.00	60.00	240.00
Rear camera (MP)	50.78	32.97	50.00	2.00	200.00
Front camera (MP)	16.60	10.87	16.00	0.30	60.00

Table 4: Summary of the six PCA inputs in the complete sample

- Summaries use **921 complete records**, before splitting (Table 4).
- Different units and ranges support **standardising before PCA**.

Outcome and feature summary

Variable	Mean	SD	Median	Min	Max
Log price	9.98	0.74	9.88	8.16	12.39
Battery capacity (mAh)	4,826.80	1,003.51	5,000.00	1,821.00	22,000.00
Screen size (inches)	6.55	0.33	6.58	3.54	8.03

Table 5: Log price and the two extra numeric predictors

Feature	Available	Phones	%
5G	No	415	45.1
5G	Yes	506	54.9
NFC	No	570	61.9
NFC	Yes	351	38.1

Table 6: Connectivity features in the complete sample

- Battery capacity (mAh) and screen size (inches) use **different scales**. Both are **standardised before regression** (Table 5).
- **5G** is available in **54.9%** of phones and **NFC** in **38.1%** (Table 6).

Why use log price?

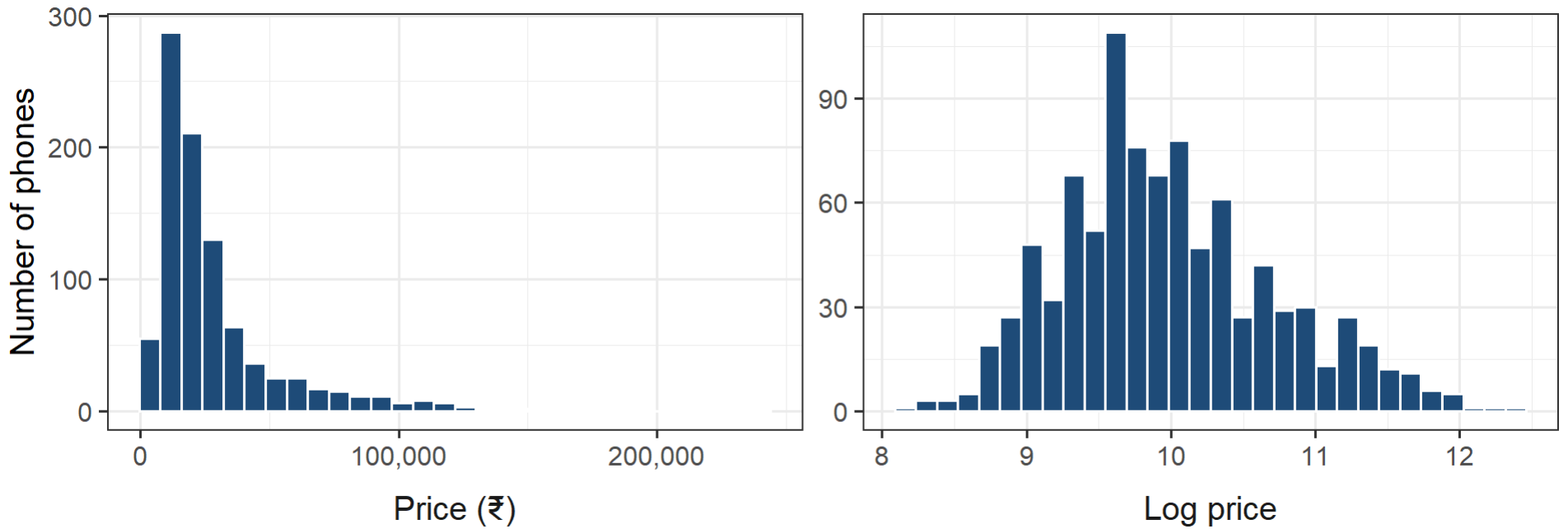


Figure 1: Price and natural log of price for the retained phones

- **921 retained phones**, matching the analysis sample (Figure 1).
- Log price **compresses the upper tail** and supports interpreting proportional price differences.

Data preparation and PCA

Are the specifications correlated?

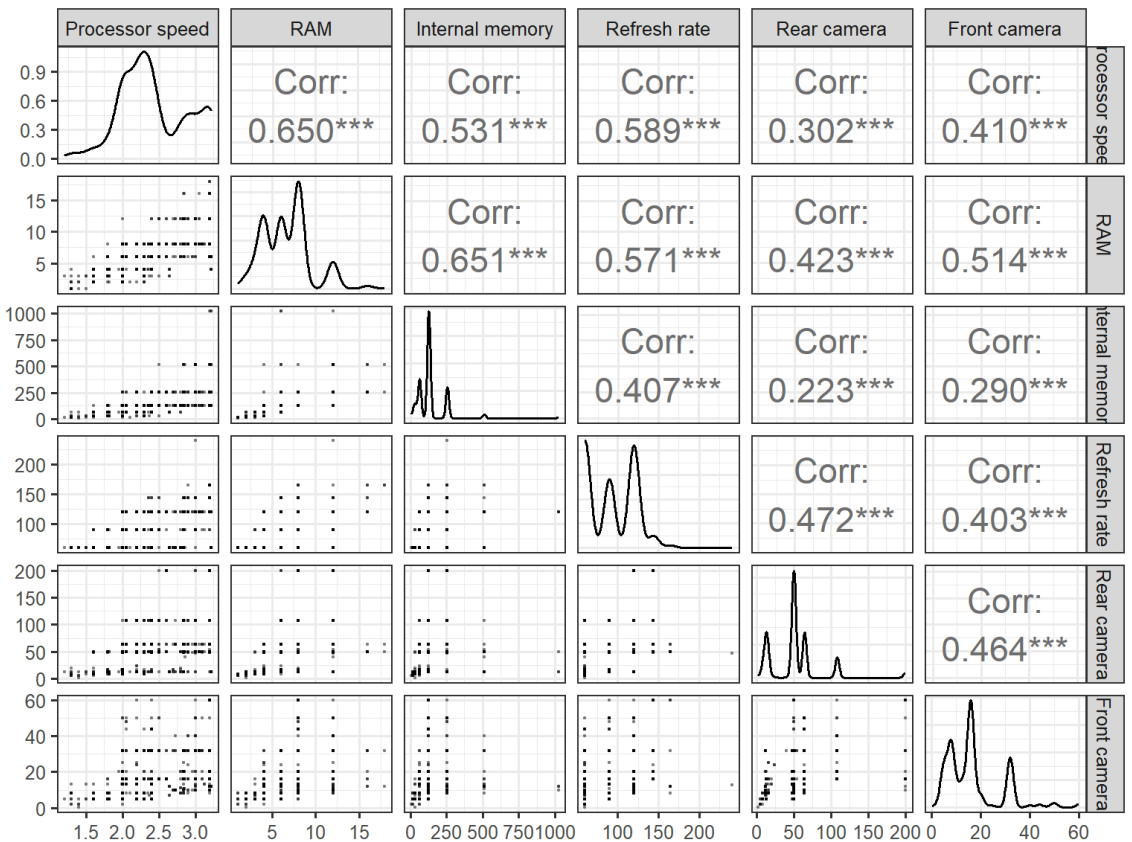


Figure 2: Correlations among the six specifications used in the PCA.

- **Correlations** range from 0.22 to 0.65, with an average of 0.46.
- **Strongest pair:** Internal memory and RAM (0.65); **weakest:** Internal memory and Rear camera (0.22).
- The specifications **overlap**, so **PCA** can combine them into **fewer** uncorrelated components.

So what

Phones can be compared on a few overall specification levels instead of six separate numbers.

Data preparation

Step	Smartphones
Original dataset	980
Luxury editions removed	2
Incomplete records removed	57
Used in the analysis	921
Training set (75%)	689
Testing set (25%)	232

Table 7: Smartphones at each preparation step.

- **921 of 980** phones used (Table 7)
- **Luxury editions** (2) removed: priced for materials, not specifications.
- **Incomplete records** removed: PCA cannot handle missing values.
- **Price** logged (strongly right-skewed); **brand** grouped into the top 7 plus Other.
- **5G** and **NFC** recoded to No/Yes; **Xiaomi** is the base brand for comparison.

So what

The results describe how specifications relate to price for typical smartphones.

Method and justification

Choice	Reason
PCA on six specifications	Specifications overlap (Figure 2); a few uncorrelated components avoid multicollinearity.
Standardise before PCA	Units differ (GB, GHz, Hz, MP); stops large numbers from dominating.
Linear regression	Log price is numeric; the question is what drives price.
Variables left out	Rating: largely explained by the specifications. Fast charging: missing for most iPhones.

Preprocessing steps

1. **Standardise** all numeric predictors.
    - Fair weighting in PCA.

2. **Combine** the six specifications into **principal components** that explain at least 70% of the variance.
    - Removes overlap and keeps key information.

3. Turn 5G, NFC and brand into 0/1 **dummy variables.**
    - Regression requires numbers; done after standardising so they stay 0/1.

4. **Drop** any **column** with a **single value**.
    - Avoids useless predictors.

Variance explained

Component	Variance (%)	Cumulative (%)
PC1	56.5	56.5
PC2	15.1	71.6
PC3	10.1	81.6
PC4	8.1	89.8
PC5	5.9	95.7
PC6	4.3	100.0

Table 8: Variance explained by each principal component.

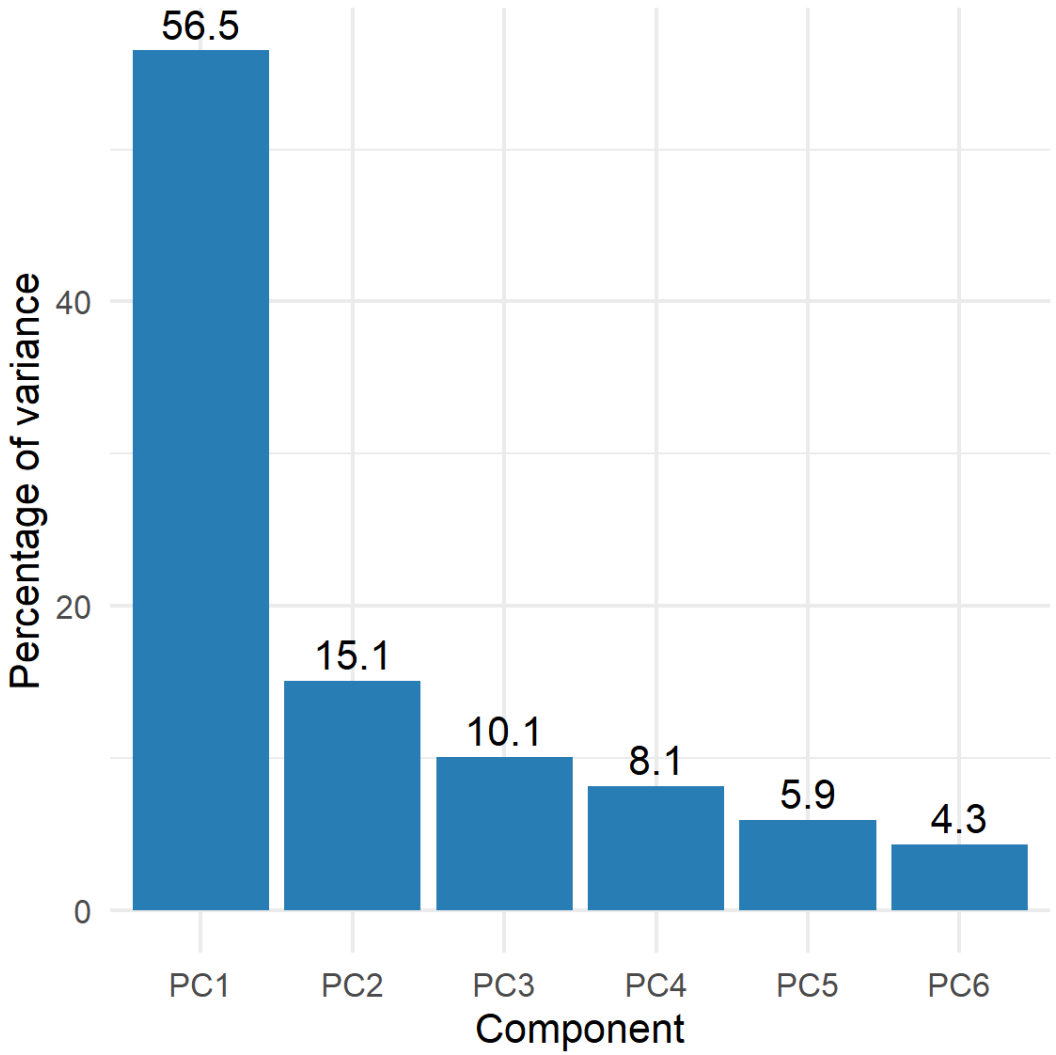


Figure 3: Percentage of variance explained by each component.

Number of components retained

- **PC1** explains **56.5%** of the variance in the six specifications (Table 8, Figure 3).
- The first **2 PCs** together explain **71.6%**, meeting the **70% threshold**.
- Each remaining component adds only **10.1% or less**.

So what

2 PCs are kept: each phone is described by 2 components instead of six specifications, with little loss of information.

What do the components mean?

How each specification loads

Specification	PC1	PC2	Kept (%)
Processor speed (GHz)	0.44	0.27	71
RAM (GB)	0.47	0.19	80
Storage (GB)	0.40	0.48	76
Refresh rate (Hz)	0.43	−0.04	62
Rear camera (MP)	0.33	−0.67	78
Front camera (MP)	0.36	−0.45	62

Table 9: Loadings on each retained component and the share of each specification they keep.

HOW TO READ THE TABLE

Loading: from −1 to 1, how strongly a specification moves with a component. Further from 0 = stronger link.

Kept: share of a specification's variation that PC1 and PC2 hold together.

- **PC1:** every loading is **positive** (Table 9)
- **PC2:** storage and processor speed **positive**, both cameras **negative**, refresh rate close to **0** (Table 9)
- **Kept:** from 62% (front camera and refresh rate) to 80% (RAM) (Table 9)

Two components describe the hardware

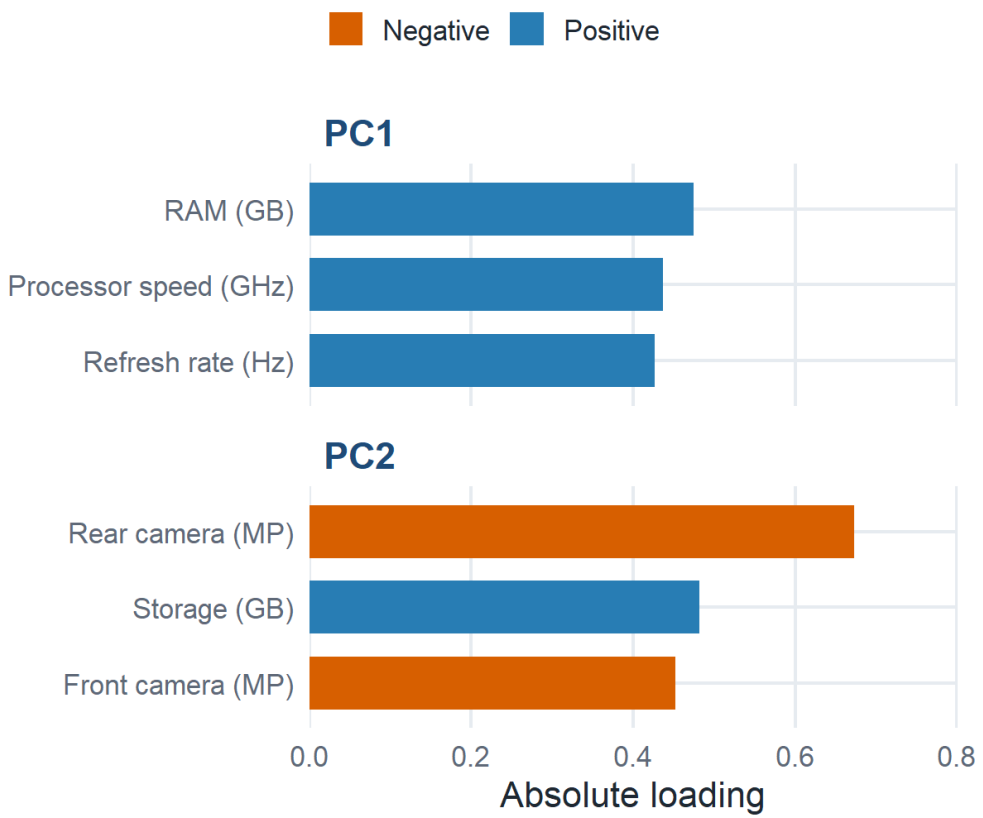


Figure 4: The three largest loadings on each component. Blue = positive, orange = negative.

PC1 · 56.5% OF THE VARIANCE

Overall performance

- All six loadings are **positive** (0.33 to 0.47), led by RAM (Table 9)
- Phones strong in one specification tend to be strong in the others (Figure 4)

PC2 · 15.1% OF THE VARIANCE

Storage and speed vs camera

- Storage (0.48) and processor speed (0.27) **positive**; rear (-0.67) and front cameras (-0.45) **negative** (Table 9)
- Refresh rate is close to 0 (-0.04), so PC2 does not reflect it (Table 9)
- A **balance**, not a level: phones lean towards memory or towards cameras (Figure 4)

What high and low PC1 scores look like

Specification (average)	Low	Middle	High
Phones	230	230	229
Processor speed (GHz)	2.03	2.35	2.89
RAM (GB)	3.9	6.6	9.1
Storage (GB)	71.3	128.6	211.3
Refresh rate (Hz)	66.3	92.4	120.7
Rear camera (MP)	27.1	55.5	69.9
Front camera (MP)	8.4	16.8	24.6
Median price (₹)	10,999	18,599	39,990

Table 10: Averages by PC1 third (training phones).

LOW VS HIGH PC1	LOW	HIGH
RAM	3.9 GB	9.1 GB
Storage	71 GB	211 GB
Refresh rate	66 Hz	121 Hz
Rear camera	27 MP	70 MP
Median price (₹)	10,999	39,990

Every specification and the median price rise from low to high: PC1 separates **budget** from **flagship** phones (Table 10).

What high and low PC2 scores look like

Specification (average)	Low	Middle	High
Phones	230	230	229
Processor speed (GHz)	2.33	2.26	2.68
RAM (GB)	6.4	5.6	7.6
Storage (GB)	120.6	108.6	181.8
Refresh rate (Hz)	96.6	87.0	95.7
Rear camera (MP)	75.8	42.4	34.1
Front camera (MP)	22.8	13.1	13.8
Median price (₹)	19,990	15,499	32,999

Table 11: Averages by PC2 third (training phones).

LOW VS HIGH PC2	LOW	HIGH
Storage	121 GB	182 GB
RAM	6.4 GB	7.6 GB
Rear camera	76 MP	34 MP
Front camera	23 MP	14 MP
Median price (₹)	19,990	32,999

A **balance**: high scorers carry more storage, low scorers bigger cameras. The middle third has the least storage (109 GB) (Table 11).

Why it matters

Two levers position a phone: **how much hardware** it has (PC1) and **where that hardware goes** (PC2).

Better-equipped phones cost more

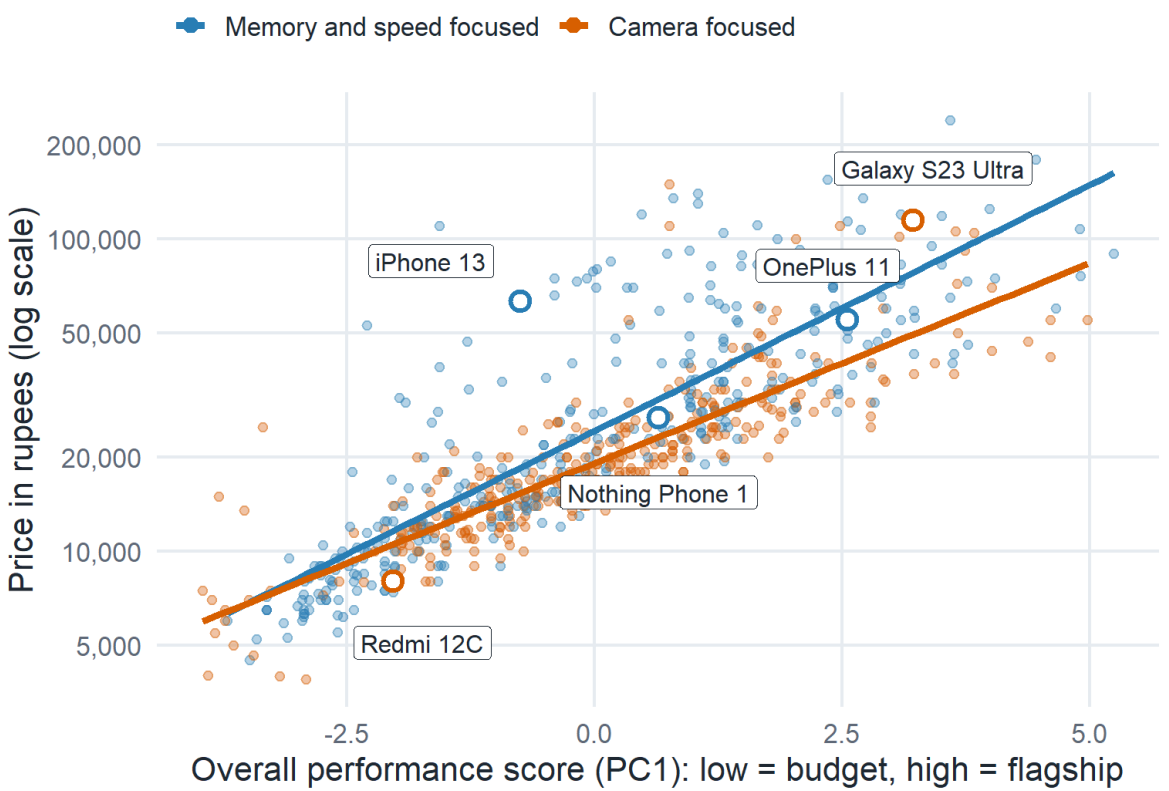


Figure 5: Price (₹) against overall performance score. Blue = leans towards memory and speed (PC2 above 0); orange = leans towards cameras (PC2 below 0). Lines show each group's trend.

0.82

CORRELATION BETWEEN PC1 AND LOG PRICE

Redmi 12C ₹7,999 to Galaxy S23 Ultra ₹114,990 (Figure 5)

- **Lines split at the top:** typical flagship ₹58,600 (memory) vs ₹39,200 (camera); typical budget phone ₹10,100 vs ₹9,400 (Figure 5)
- **iPhone 13:** ₹62,999, against ₹14,750 for phones with a similar PC1 score (Figure 5)

The memory premium is largest among flagships

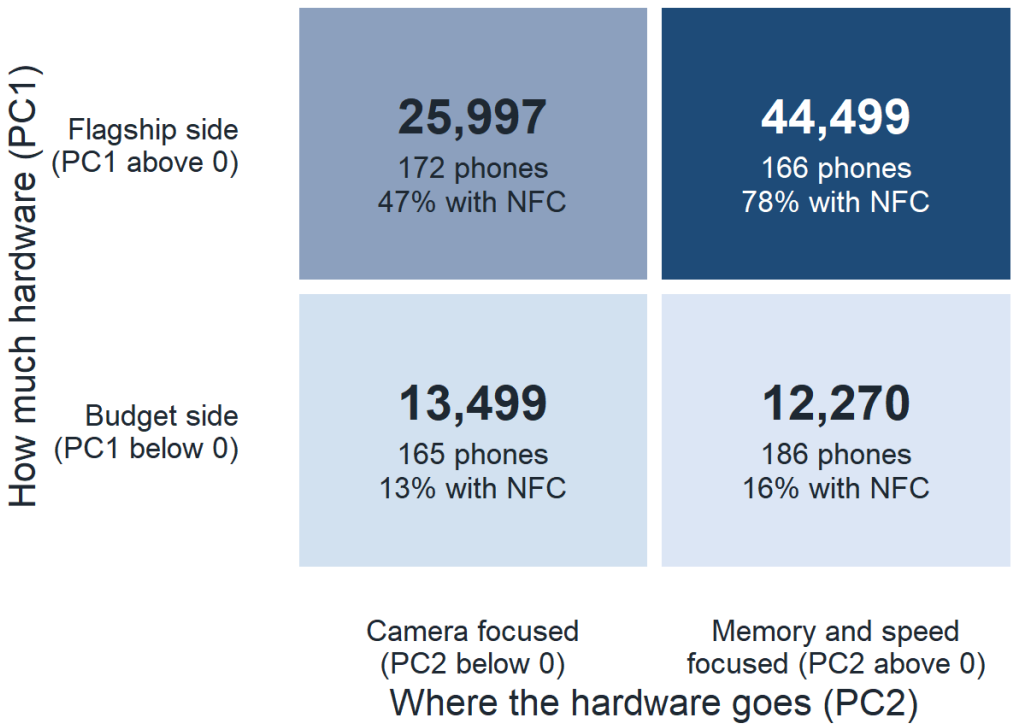


Figure 6: Median price (₹) of the training phones in each corner of the PC1 × PC2 map (split at 0 on each score).

- **Flagships:** memory ₹44,499 vs camera ₹25,997 (Figure 6)
- **Budget phones:** no clear premium; camera ₹13,499 vs memory ₹12,270 (Figure 6)
- **NFC:** 78% of memory-focused flagships vs 47% of camera-focused ones (Figure 6)

So what

Storage and speed go with the highest prices **at the top end**; budget phones can lead on cameras at a similar price.

Regression Results

What drives smartphone prices?

Dependent variable: **log price**. Base categories: **Xiaomi**, **no 5G**, **no NFC**. We read the **significance level** and the **sign** only, not the size of each coefficient.

Term	Coefficient	p-value	Significance
Intercept	9.775	<0.001	0.1%
Battery capacity	-0.120	<0.001	0.1%
Screen size	0.099	<0.001	0.1%
PC1: overall performance	0.262	<0.001	0.1%
PC2: storage and speed vs camera	0.100	<0.001	0.1%
5G (vs no 5G)	0.077	0.0150	5%
NFC (vs no NFC)	0.261	<0.001	0.1%

Table 12: Components, specifications and features

Brand	Coefficient	p-value	Significance
Apple	0.904	<0.001	0.1%
Motorola	-0.149	0.0097	1%
Oppo	0.064	0.1644	Not significant
Realme	-0.027	0.5649	Not significant
Samsung	0.245	<0.001	0.1%
Vivo	0.126	0.0065	1%
Other brands	0.006	0.8764	Not significant

Table 13: Brands compared with Xiaomi

Specifications vs price

- **Overall performance (PC1): positive significant influence** on price at the **0.1% level** (Table 12); stronger all-round phones are priced higher.
- **Storage and speed vs camera (PC2): positive significant influence** at the **0.1% level**; storage- and speed-focused phones are priced higher than camera-focused ones.
- **Screen size: positive significant influence** at the **0.1% level**; larger screens, higher price.
- **Battery capacity: negative significant influence** at the **0.1% level**; larger batteries, lower price.

So what

Price rises with the overall performance bundle and screen size; bigger batteries are linked to lower prices, and extra camera megapixels alone do not raise price.

Features and brands vs price

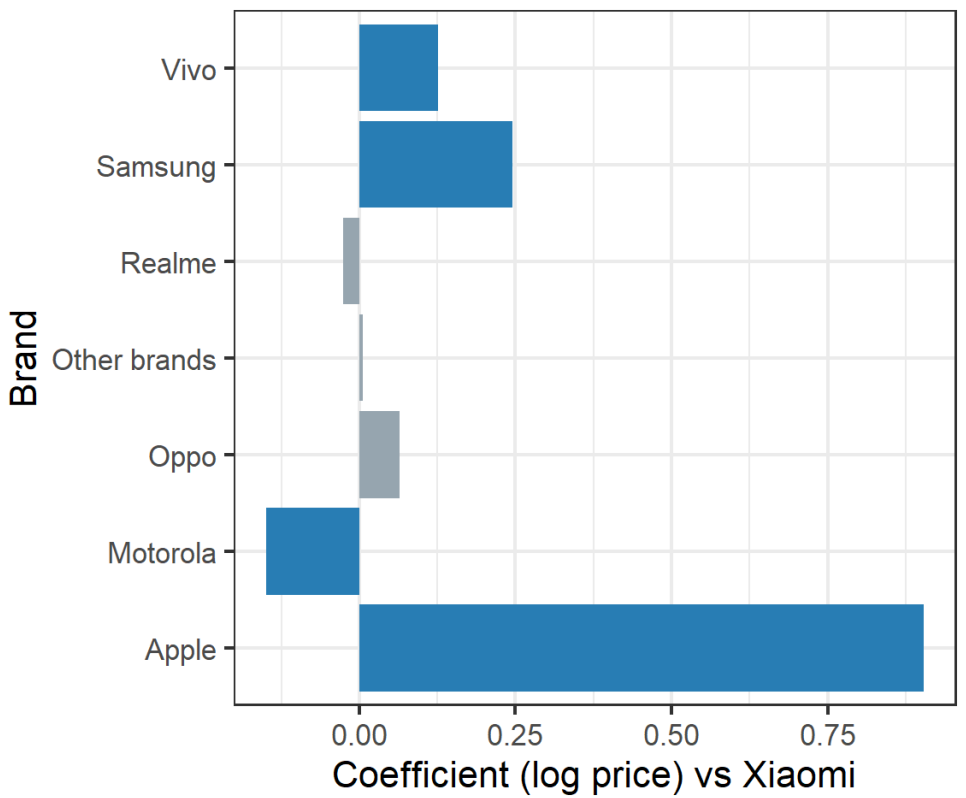


Figure 7: Brand coefficients compared with Xiaomi (blue = significant at 5%, grey = not significant)

- Phones **with NFC:** significantly **higher** price than without NFC (0.1% level).
- Phones **with 5G:** significantly **higher** price than without 5G (5% level).
- **Apple, Samsung** (0.1%) and **Vivo** (1%): significantly **higher** than Xiaomi (Table 13, Figure 7).
- **Motorola:** significantly **lower** than Xiaomi (1% level).
- **Oppo, Realme** and **other brands: not significantly different** from Xiaomi.

Summary

Key findings

1 · PCA

Two components summarise the hardware

Overall performance (PC1) and storage and speed vs camera (PC2) explain 71.6% of the variation in the six specifications (Table 9, Figure 4)

Use: compare a phone with its rivals on 2 scores instead of 6 specifications

2 · SPECIFICATIONS

Better-equipped phones cost more

Both components and screen size raise price at the 0.1% level; a larger battery lowers it (Table 12)

Size: median ₹10,999 (low PC1 third) vs ₹39,990 (high third) (Table 10)

3 · CONNECTIVITY

NFC and 5G add a premium

NFC at the 0.1% level and 5G at the 5% level, compared with phones without them (Table 12)

Where: NFC in 78% of memory-focused flagships vs 13% of camera-focused budget phones (Figure 6)

4 · BRAND

Brand matters beyond hardware

Apple, Samsung and Vivo are priced above Xiaomi, Motorola below; Oppo, Realme and Other do not differ (Table 13)

Example: iPhone 13 at ₹62,999 vs ₹14,750 for phones with a similar PC1 score (Figure 5)

Bottom line

Price follows the **whole hardware bundle** and the **brand name**. The memory premium is largest among flagships (₹44,499 vs ₹25,997), with no clear gap among budget phones (Figure 6, Table 12).

Recommendations

Recommendation 1: performance

Finding

Overall performance (PC1) and the storage and speed focus (PC2) both have a **positive significant influence** on price at the 0.1% level, while battery capacity has a **negative** one (Table 12).

Action for manufacturers

- To launch a **higher-priced model**, upgrade the **whole performance bundle**: processor, RAM, storage, refresh rate and cameras together.
- At the **same overall performance**, favour **storage and processing** over extra camera megapixels: storage- and speed-focused phones are priced higher than camera-focused ones, most of all among flagships (Figure 6).
- Use **large batteries** to add value to **budget models**, not to charge more.

Recommendation 2: brand gap

Finding

With similar specifications, **Apple, Samsung and Vivo** are priced significantly **higher** than Xiaomi, while **Oppo and Realme** are not significantly different and **Motorola** is **lower** (Table 13, Figure 7).

Action for retailers and brands

- Display **Xiaomi** phones next to **Samsung and Vivo** models with matching specifications, and promote them as **“same specifications, lower price”**.
- **Motorola**, priced below Xiaomi, should compete on **specifications per price** rather than raise prices.

References

- Abdur Rahman. (n.d.). *Smartphone Models Dataset*. Kaggle. Local file: `smartphone_cleaned_v5.csv`.
- ETW2500 teaching materials: *Principal Component Analysis* (`PCA.pdf`), `tutPCA.qmd`, descriptive-analysis notes, and `tut tidymodels reg student.qmd`.
- Wickham, H., et al. (2019). *Welcome to the tidyverse*. Journal of Open Source Software, 4(43), 1686.

Thank you

Questions?